A Comparison of Volatility-Managed Portfolio Models

Team 02

Niels Blom	Patrick Huizinga	Linda Li	Lajla Mutabdzija
617282	613051	619601	590106

June 2023

Word count: 4817
Abstract word count: 138

Abstract

This paper employs multiple models that predict the volatility of shocks today based on past information. Besides the well-known GARCH model, we introduce alternative models that deal with asymmetry in returns, fat-tailed shock distributions and inter-dependencies between different assets. All models are used to create volatility-managed portfolios based on forecast volatility. We analyze the performance of the different portfolios by means of five pre-constructed Fama-French factors over 60 years. We use different metrics to compare our models to two benchmark models. We find that most models are able to yield superior results over the benchmark models based on our performance metrics. A simple equal weight portfolio still yields the highest annual returns. But we conclude that the multivariate Dynamic Conditional Correlation (DCC) model provides the best risk-adjusted performance and lowest skewness, and is therefore the optimal choice for investors.

1 Introduction

Portfolio optimization has been a part of finance research for decades. In this field, an investor tries to select the best possible portfolio according to some objective. Most often, this means maximizing the expected return of a portfolio whilst minimizing its financial risks. The usage of developed theory on portfolio management can improve the risk-return characteristics of a portfolio [Markowitz, 1991]. The developed portfolio theory is therefore of interest to any rational investor aiming to maximize their profits, given a level of risk.

The simplest investment strategy is the 1/N portfolio, which allocates equal weights to all assets in a portfolio. The strategy does not make a distinction between assets. In other words, it invests an equal share of wealth in each asset available, regardless of its characteristics. Although DeMiguel and Garlappi [2009] has shown that this strategy performs surprisingly well under various market conditions, other studies suggest that it could be improved by managing the volatility of the portfolio [DeMiguel et al., 2013].

To understand why an investor should be interested in the volatility of an asset, one should carefully look at how these financial instruments behave. The asset returns in a portfolio follows some kind of trend, but swing around that trend over time. That is, the returns of that asset are on average higher in some periods, whilst being lower in others. The dynamic features of the asset market cause the deviation around the trend to be non-constant. In other words, the asset returns usually vary over time and are often heteroskedastic. If an 1/N model is used, it could very well be that in periods of high volatility, the risk taken by the investor is higher than preferred.

The main goal of this paper is to investigate whether asset allocation weighted by forecast volatility, performs better than two benchmark models. And if so, which model performs best. The first benchmark model is the equal weight 1/N portfolio. The second benchmark model is the model proposed by Moreira and Muir [2017], which aims to manage the risk of a portfolio by using weights inversely proportional to the realized variance of the previous month. These benchmark models are compared with several others models.

The models we implement in this paper are the Generalized ARCH (GARCH), the Exponential GARCH (EGARCH), Generalized Auto-regressive Score (GAS) model and the multivariate Dynamic Conditional Correlation model (DCC). We will also introduce GARCH models which try to model the behavior of the error terms in a more realistic way, by introducing different shock distributions. All these models try to, in some way, create a volatility-managed portfolio by forecasting the variance of different assets in the future, based on information available up till today. These forecast variances are then used to assign weights to the different assets in each period, for which a portfolio is constructed.

We use 5 pre-constructed Fama-French factors to setup portfolios and test the performance of the models. To compare the performance of different portfolios, we use multiple metrics such as the annualized return, the Sharpe and Sortino ratios and skewness of the portfolio. These metrics give measure to both the risk and the return of the constructed portfolio.

We find that of both the benchmarks, the 1/N model provides the highest annualized return. But, if we consider risk-adjusted portfolio performance, we find that the simple volatility managed benchmark portfolio performs better. We found that for the GARCH model, for different one-step ahead forecasts with a rolling window, we were able to provide the best metrics using windows of 240 months. The forecasting window of 240 months is then used for all GARCH based models.

We did not find a model that yields a higher annual return than the 1/N model, but a large part of the models were able to provide superior risk-adjusted metrics performances over the benchmark models. The multivariate DCC model was able to provide the best Sharpe and Sortino ratio. Also, the model provides a very low skewness, indicating relatively balanced returns.

This paper is further structured as follows. After a literature review, the available data is presented in Section 2 and the Benchmark models are explained in Section 3. The models used to attempt to beat the Benchmark models, are thoroughly explained in Section 4. The performance of these models are given in Section 5, where we also provide a basis for our window size selection, and the paper is concluded in Section 6.

Literature Review

The practice of introducing a mathematical framework for constructing an optimal portfolio which maximizes returns, given a level of risk, was introduced by Markowitz [1952]. The key finding of this paper was that an investor could improve their portfolio performance by incorporating volatility of assets in investment decisions. Diversified portfolios can reduce the risk of an investor, whilst not affecting the expected returns. This established a foundation for future research on volatility in portfolio theory. Econometric models could provide key insights in how one can predict, and incorporate, future volatility in a model.

One of the key econometric models that was developed was the autoregressive conditional heteroskedasticity (ARCH) model [Engle, 1982]. The model describes the volatility of a shock today based on the previous squared error terms. An ARCH model can be estimated using ordinary least squares. For multiple years, this type of model was widely used in finance and econometrics to forecast time-varying volatility and clustering volatility of portfolios. Bollerslev [1986] built upon this by not only including past squared error terms in the variance equation, but also including past error variances. This Generalized ARCH (GARCH) model allowed for modelling of more complex volatility patterns. A downside is that this model is generally harder to estimate. Instead of OLS, used by the ARCH model, a maximum likelihood approach is employed to obtain parameter estimates. The GARCH model can improve portfolio performance, as shown by Nugroho et al. [2019].

The GARCH model however also has limitations. For example, it is known that the GARCH model, due to its symmetric distributions, fails to model the leverage effect of volatility often observed in financial data. Many extensions have been proposed as a solution, such as the Exponential GARCH (EGARCH) model [Nelson, 1991], the Glosten-Jagannathan-Runkle GARCH (GJR-GARCH) model [Glosten et al., 1993], and the Asymmetric Power ARCH (APARCH) model [Ding et al., 1993]. The difference between these models lie in the way the asymmetric effect is modelled. Usually, the conditional variance is modified to include terms for the asymmetry. We selected EGARCH because it seems to be preferred in the different settings of Alberg et al. [2008] and Rodríguez and Ruiz [2012].

Another point that should be addressed on the GARCH model is the potential inability to deliver accurate volatility forecasting after the occurrence of a large shock in the market. As will be discussed more in-depth in Section 4.3, we suspect that the distribution of shocks are fat-tailed, which implies that shocks may sometimes be exceedingly large and require specific modelling attention. An alternative to GARCH which can address this issue is the Generalized Auto-regressive (GAS) model [Creal et al., 2013].

Another property of GARCH is that it does not account for the correlation between different assets in a portfolio, as GARCH is a univariate model. Bollerslev [1990] first advocated the usage of a multivariate GARCH model. Caporin and McAleer [2014] made a large comparison for robustness of different multivariate GARCH models. It used a one-step ahead rolling window to compare the out-of-sample forecasting performance for the different models. They concluded that there is no real optimal model and that usefulness of different models is largely dependent on the data and sample period implemented in the model, but in general one should prefer the accuracy and flexibility of the Dynamic Conditional Correlation (DCC) model.

In this paper, we compare EGARCH, GAS and DCC models to the 1/N model, adding further insights to the literature.

2 Data

In this section, the data used will be discussed and explained. For the ready-made portfolio, we have used the daily returns from July 1963 up till March 2023 for the five Fama-French factors from [French Data Library]. Note that all factors consider returns in percentages.

The factors are: Excess market return ($Mkt - RF$: Market minus risk-free) calculated by subtracting the risk-free rate from the the market rate, secondly the Size of the firms (SMB: Small minus big) which is the excess return of a portfolio of small-cap stocks over a portfolio of large-cap stocks, thirdly the Book-to-Market Value (HML: High minus low) which is the excess return of a portfolio of high book-to-market value stocks over a portfolio with a low one, next the Momentum Profitability (RMW: Robust minus weak) capturing the excess return portfolios of stock with robust momentum over portfolios containing

stocks that have a weak momentum, and finally Investment (CMA: Conservative minus aggressive) which captures the excess return of portfolios of companies with low investment strategies over portfolios of companies with high investment strategies.

The first 3 factors are Fama-French factors which were introduced in Fama and French [1996], the last two are expansions from Fama and French [2015]. Besides that, due to the reason that an investor must borrow money to invest in the market, the risk-free rate is already subtracted from the first factor. The other four factors are long-short and do not need any initial investment, hence the risk-free rate is not subtracted.

To gain more insight into the factors, Table 1 is provided with descriptive statistics for each individual factor at hand. Each column represents one of the market factors as described above. In this table we include the *Annualized returns* for each factor to summarize the performance of the individual factors.

The first factor ($Mkt - RF$) has the highest mean daily return and annual return percentage among the factors (as seen from the difference between Maximum and Minimum values). However, the standard deviation is also the highest, and the range of observations is the widest among the factors, which indicates higher volatility. The worst performing portfolio seems to be SMB. It has the third highest standard deviation, and has the second largest widest range of observations, while it has the lowest mean and annual return percentage. The other factors seem to be more balanced in terms of risk and return.

All factors have a high kurtosis and skewness, implying the presence of fat tails and outliers in the error term distribution. Normality for any of the five market factors is rejected by the Jarque-Bera test statistic. It does not seem likely that the shocks follow a normal distribution. This has implications for GARCH, and will be discussed more in-depth in Section 4.3.

Table 1: Descriptive Statistics of the 5 Fama-French Factors

	Mkt-RF	SMB	HML	RMW	CMA
Mean	0.027	0.007	0.014	0.014	0.014
Ann.Ret (%)	5.58	1.48	3.26	3.31	3.27
Maximum	11.35	6.17	6.74	4.52	2.53
Minimum	−17.44	−11.19	−5.00	−3.01	−5.87
Std. Dev.	1.03	0.54	0.58	0.40	0.378
Skewness	−0.53	−0.70	0.28	0.34	−0.34
Kurtosis	18.56	21.22	14.24	12.28	12.41
Jarque-Bera	152503	209232	79361	54226	55838

3 Benchmark Models

In this section, two benchmark models are employed to measure the performance of theoretical portfolio construction methods. Both models will be briefly described and details on asset weight calculations are provided. The benchmark model performances will be used for future comparison.

The simple volatility managed portfolio re-balances each month proportionally to the inverse of the factor's variance of the previous month. This method allows us to find and assign weights to assets based on the forecast variances. As this weight calculation will be used extensively in this paper, we will briefly introduce it. A more extensive description is given by Moreira and Muir [2017].

Let t be the index count for the months, and let the 1-day-ahead forecast of the predicted variance for an asset i be given by $(\hat{\sigma}^{(i)}_{t+1|t})^2$ where the conditional bar indicates that information for the prediction is available up till time t. If we have $(\hat{\sigma}^{(i)}_{t+1|t})^2$ at hand for each asset $i = 1, \ldots, N$ in a portfolio, we determine the associated weights for an asset i at time t by:

$$w_t^{(i)} = \frac{1}{(\hat{\sigma}_{t+1|t}^{(i)})^2} \left[\sum_{i=1}^{N} \frac{1}{(\hat{\sigma}_{t+1|t}^{(i)})^2} \right]^{-1}. \tag{1}$$

Note that the right term ensures that the calculated weights for all assets sum up to one. Because of this, a combination of assets is chosen such that the investor's wealth is fully invested.

In Table 2, annualized return percentages are given for portfolios based on each of the 5 introduced factors, as well as the two benchmarks discussed earlier: the 1/N portfolio and the simple volatility managed portfolio. The Sharpe ratio, Sortino ratio and skewness are calculated with the monthly returns. Note that the Sharpe and the Sortino ratio characterize the ratio of the returns compared to the standard deviation of those returns. Compared to the Sharpe ratio, the Sortino ratio does not penalize returns above the mean, something we feel few will object to. Note that the skewness measures the asymmetry in the factor returns. It therefore tells us about the balance between the occurrence of negative and positive returns.

The first five portfolios invest wealth in a single factor. The 1/N portfolio invests 20% in each of the five factors. The re-balancing of the portfolio occurs at the start of each month. Furthermore, the volatility-managed portfolio assigns weights to each asset, at the start of every month, based on the forecast variance for the upcoming month. For the benchmark model, we follow the approach by Moreira and Muir [2017], where the realized daily variance S_t^2 of an asset is used as $\hat{\sigma}_{t+1}^2$.

Table 2: Benchmark Comparisons

	Mkt-RF	SMB	HML	RMW	CMA	1/N	Simple Vol.
Ann. Returns (%)	5.58	1.48	3.26	3.31	3.27	3.85	3.25
Sharpe Ratio	0.43	0.19	0.36	0.46	0.49	0.86	0.87
Sortino Ratio	0.59	0.30	0.57	0.69	0.87	1.16	1.36
Skewness	-0.50	0.18	0.24	0.41	0.55	-0.29	0.29

The simple full market portfolio performs the best in terms of annualized returns, with the 1/N portfolio providing the second best results. This measure accounts for the investor's returns, but does not account for their exposure to risk. In terms of Sharpe and Sortino ratios, which do account for the risk taken by the investor, the simple volatility managed portfolio yields the best performance, with again the 1/N portfolio ranking second. The 1/N portfolio ranking second on the Sharpe and Sortino ratios, thereby outperforming the individual market factors, provides an empirical argument for the convention that diversification leads to a better portfolio risk management. Both benchmark models provide a relatively low skewness compared to the individual market factors. The difference in Sharpe ratio between the 1/N and simple volatility managed portfolio is minimal, whilst the Sortino ratio differs by about 20%. This suggests that while the variance of both portfolios are equal proportionally to their return, the 1/N portfolio has, on average, a larger exposure to downside risk.

Both benchmark models will be used in Section 5 for comparison with different introduced models.

4 Models

4.1 GARCH Model

In this section, a description will be given on the GARCH model of order (1,1), introduced by Bollerslev [1986].

The GARCH model is a statistical tool for estimating the future volatility of a financial asset. It is a time-series model capable of capturing the persistence and time-varying aspect of volatility in financial data.

We first introduce a general dynamic model which captures the time-varying volatility of asset returns. Note that the subscript (i) for asset i is suppressed for notational convenience. The daily return r_t of an asset is described by:

$$r_t = \mu + \sigma_t \varepsilon_t, \qquad 1 \leq t \leq T. \tag{2}$$

For now, it is assumed that the shocks $\varepsilon_t \overset{iid}{\sim} \mathcal{N}(0,1)$. Furthermore, T denotes the sample size of the return series. If σ_t and ε_t are known, it holds that $Var(r_t) = \sigma_t^2$. But in practice ε_t and σ_t are not observed separately and hence this framework is not sufficient for modelling the volatility of assets.

For prediction purposes, it is necessary to introduce a filter; a general rule on how σ_t can be forecast. The model for the GARCH(1,1) filter can be described as:

$$\hat{\sigma}_{t+1}^2 = \omega + \alpha \hat{\sigma}_t^2 \hat{\varepsilon}_t^2 + \beta \hat{\sigma}_t^2, \tag{3}$$

where α controls the impact of previous shocks, and β controls the persistence, or memory, of the volatility estimate. Note that the entire right-side of Equation (3) can be evaluated at time t. Hence $\hat{\sigma}_{t+1}^2$ can be predicted at time t. We impose the parameter restrictions $\alpha \geq 0$, $\beta \geq 0$ and $\omega \geq 0$ to ensure that $\hat{\sigma}_t^2 \geq 0$ for any t. Also, $\alpha + \beta < 1$ is imposed to ensure mean-reverting of $\hat{\sigma}_t^2$ towards the unconditional expected variance.

The non-linear nature of this GARCH filter ensures the usage of maximum likelihood estimation for the parameters $\boldsymbol{\theta} = (\mu, \omega, \alpha, \beta)$. Under the assumption of normally distributed error terms, this parameter vector is found by solving:

$$\hat{\boldsymbol{\theta}}_{ML} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, \mathcal{L}(\boldsymbol{\theta}) = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \prod_{t=1}^{T} \left[\frac{1}{\sqrt{2\pi\hat{\sigma}_t^2}} \exp\left\{ -\frac{1}{2} \frac{(r_t - \mu)^2}{\hat{\sigma}_t^2} \right\} \right], \tag{4}$$

where T represents the number of periods, $\hat{\sigma}^2$ is the estimated variance of the error term, and r_t are the returns in period t. Note that there is no closed-form solution available for this maximization. A numerical optimization procedure is used to obtain $\hat{\boldsymbol{\theta}}_{ML}$.

After the variances $\hat{\sigma}_t^2$ are forecast for each period $t = 1, \ldots, T$, the asset weights w_t are computed similarly to that of the simple volatility managed benchmark model, as described in Equation (1).

To account for excess kurtosis of the data, we will also test variants of the GARCH model that assume the shocks are distributed according to the Student's t and the Generalized Error Distribution (GED). In both cases ν (and λ) are part of $\boldsymbol{\theta}_{ML}$.

4.2 Exponential GARCH Model

In many cases, asymmetry in stock market return series are detected in the data sets. However, the GARCH model assumes that positive and negative shocks have a symmetric effect on volatility of asset returns. When there are signs of non-linearity and asymmetric impact on the volatility of financial returns, non-linear GARCH models are employed to manage volatility. This implies that the response of volatility to market shocks depends on both the direction and magnitude of the shock. In other words, the model takes into account that negative shocks may influence future volatility in a different way than positive shocks do.

In Table 2, we observe that the skewness differs from zero for both benchmark models, which leads to a suspicion of possible asymmetry in the factors. Also, the minimum and maximum returns of the different factors in Table 1 differ, indicating greater negative outliers compared to positive outliers in the factor returns. A flexible, empirically tested model that captures these assymetric effects in volatility is the EGARCH model.

EGARCH provides a different filter than the standard GARCH model. This filter predicts $\hat{\sigma}_{t+1}^2$ at time t as follows:

$$\log\big(\hat{\sigma}_{t+1}^2\big) = \omega + \alpha\big(|\hat{\varepsilon}_t/\hat{\sigma}_t| - \gamma \cdot \mathbb{E}(|\hat{\varepsilon}_t/\hat{\sigma}_t|)\big) + \beta \log\big(\hat{\sigma}_t^2\big). \tag{5}$$

In this model, $\log(\hat{\sigma}^2_{t+1})$ represents the (logarithmic) conditional variance at time $t+1$. ω represents an intercept term, α is the coefficient which captures the impact of past standardized shocks on the future volatility. β is a coefficient that captures the persistence of conditional variance over time t, and γ is the parameter allowing for asymmetry. Note that in this method, it is assumed that z_t approximately follows a standard Gaussian distribution.

Different from the standard GARCH model, there are no restrictions on the parameters to allow for non-negativity of the conditional log variances. The asymmetry is influenced by the parameter γ in the filter. If γ is positive, a positive return shock would have the same impact on volatility as a negative return shock of the same magnitude, meaning positive and negative shocks have symmetric effects on volatility. Conversely, if γ is negative, a positive return shock will have a smaller impact on volatility compared to a negative return shock of the same magnitude.

Notably, by taking the logarithm of the conditional variance, allows for random oscillatory behaviour in the volatility process [Nelson, 1991].

The non-linear nature of this model leads to a maximum likelihood approach estimate the corresponding model parameters. One can obtain parameter estimates for $\hat{\boldsymbol{\theta}}_{ML} = (\omega, \gamma, \alpha, \beta)$ by the log-likelihood maximization underneath. Like with GARCH, there is no closed-form analytical solution available, and we thus require a numerical procedure.

Note that we again compute the weights w_t for $t = 1, \ldots, T$ similar to Equation (1). We also test variants of EGARCH that assume the shocks are distributed as Student's t and GED, like for GARCH.

4.3 GAS Model

The Generalizes Auto-regressive Score (GAS) Model allows for a large set of model specifications. It allows a user to model and predict time-varying parameters, such as volatility of returns. We will give a small summary of the general model, using the notation of Creal et al. [2013].

Denote some time-varying parameter f_t and some observable y_t and let the conditional density of the observable be $p(y_t|f_t)$. Then f_{t+1} can, at time t, be described by the recursion:

$$f_{t+1} = \omega + \beta f_t + \alpha S(f_t) \left[\frac{\partial \ln p(y_t|f_t)}{\partial f_t} \right], \tag{6}$$

where $S(f_t)$ denotes some scaling function of the score of $\ln p(y_t|f_t)$. The score measures the sensitivity of $\ln p(y_t|f_t)$ with respect to changes in parameters. A more detailed mathematical description of the general model can be found in Creal et al. [2013].

In our application, this time-varying parameter $f_t = \hat{\sigma}^2_t$ and $y_t = r_t - \mu$. Certain choices of parameters, distributions and score functions in Equation (6) can provide a wide class of model specification for time-varying parameters, such as the discussed GARCH(1,1) model.

Using $Y_t \sim \mathcal{N}(0, f_t)$ gives:

$$\begin{aligned} \frac{\partial \ln p(y_t|f_t)}{\partial f_t} &= \frac{y_t^2 - ft}{2f_t^2}, \\ S(f_t) &= 2f_t^2, \\ f_{t+1} &= \omega + \alpha(y_t^2 - f_t) + \beta f_t, \end{aligned}$$

which is equivalent to Equation (3) that defined the GARCH(1,1) model [Creal et al., 2013].

As noted in Section 2, we suspect that the factors at hand are not normally distributed. All factors have a relatively high kurtosis, indicating their distribution is fat-tailed, and occurrences of outliers are relatively likely. Motivated by these findings, we model the shocks as a Student's t distribution. Motivated by the GAS model description above, we change the GARCH(1,1) filter to a filter that incorporates this Student's t shock distribution. Specifically for the Student's t distribution, this scoring rule takes into account the degrees of freedom that controls the fatness of the tail of the distribution. The GAS model provides the following filter after providing the correct parameters to Equation (6):

$$f_{t+1} = \omega + \alpha(1 + 3\nu^{-1}) \left[\frac{1 + \nu^{-1}}{(1 - 2\nu^{-1})(1 + \nu^{-1} y_t^2/(1 - 2\nu^{-1}) f_t)} y_t^2 - f_t \right] + \beta f_t, \tag{7}$$

where ν represents the degrees of freedom of the shock distribution.

To obtain parameters $\boldsymbol{\theta} = (\omega, \alpha, \beta)$ for this observation-driven model, we again lean on a maximum likelihood approach. The GAS model assumes that the parameters are time-varying and follow a general auto-regressive stochastic process. With the assumption on the distribution of shocks following a Student's t distribution, and with the inclusion of time-varying parameters, we can again not provide a closed-form solution for $\boldsymbol{\theta}$. For a detailed description on how one can estimate the parameters in the above model, we refer to Creal et al. [2013], which provide an iterative estimation procedure. The required weights w_t for $t = 1, \ldots, T$ are determined as in Equation (1).

4.4 Multivariate GARCH Model - DCC

Whilst the standard GARCH models account for the time-varying volatility of single asset returns, it does not account for any correlation between different assets. Section 4.4 shows the covariance matrix estimation between the 5 factors.

Table 3: Covariance Matrix Estimation

	Mkt-RF	SMB	HML	RMW	CMA
Mkt-RF	1.05	−0.038	−0.10	−0.085	−0.14
SMB	−0.038	0.30	0.04	−0.06	0.00
HML	−0.10	0.04	0.34	0.01	0.12
RMW	−0.08	−0.06	0.01	0.16	0.02
CMA	−0.14	0.00	0.12	0.02	0.14

As observed above, the covariance between the different factors differ from zero. Relative to the variance of the assets on the diagonal, the covariance estimations show that inter-dependencies between factors should certainly not be neglected.

To account for the inter-dependencies, and model possible dynamic correlation over time between factors, we will introduce a convenient multivariate GARCH model. The Dynamic Conditional Correlation (DCC) model was first introduced by Engle [2002]. The model estimates univariate GARCH(1,1) models for each asset in a portfolio. Then, it estimates DCC parameters for which a time-varying covariance matrix can be estimated. This matrix captures the possible time-varying dependencies between assets.

Denote H_t the conditional covariance matrix of a N dimensional time series y_t. The model is then:

$$H_t = D_t R_t D_t, \tag{8}$$

$$R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}, \tag{9}$$

$$Q_t = S(1 - \alpha - \beta) + \alpha \varepsilon_{t-1} \varepsilon'_{t-1} + \beta Q_{t-1}, \tag{10}$$

where D_t is a diagonal matrix where the square root of the estimated univariate GARCH variances is placed on the diagonal. The matrix R_t represents the time-varying conditional correlations. These are based on $\text{diag}(Q_t)$, another diagonal matrix with the same diagonal elements as Q_t. This matrix, in turn, is derived from matrix S, which is the sample correlation matrix of ε and is a consistent estimator of the unconditional correlation matrix. α and β are positive parameters and $\varepsilon_t = D_t^{-1} y_t$ are the standardized but correlated residuals, meaning the conditional variances of the components of ε_t are equal to 1, but the conditional correlations are given by R_t. One is also able to show that R_t is the conditional covariance matrix of ε_t.

The parameters are estimated by a three-step quasi-maximum likelihood (QML). For the three likelihood functions, we refer to section 2 of Hafner and Reznikova [2012]. First, parameters for the volatility process are estimated. These are needed to compute the moment estimator $\hat{S}$ of S. Finally, we are able to estimate the parameters of the correlation process. The resulting matrix H_t is then incorporated into the GARCH framework through its likelihood function. This function is constructed based on the assumption that the asset returns follow a multivariate normal distribution with mean zero and time-varying covariance matrix H_t. The GARCH parameters are then estimated as usual.

5 Results

This section starts of by testing the performance of GARCH using several window sizes. Later Section 5.2 shows and discusses the performance of all aforementioned models.

5.1 Window Selection

All GARCH-like models depend on a window of historical data to estimate the parameters. To make sure we select the right window size, we first tested several rolling window sizes on the GARCH(1,1) model.

Furthermore, we use a monthly and daily rolling window with window sizes $W_m \in [48, 120, 240]$ months and $W_d \in [1000, 2500, 5000]$ days. For this rolling window estimation we choose a fixed window of observations, which is some fixed sample size we use to estimate the model. With the estimated model, we forecast the variance of the next month. This is repeatedly done, such that for each following month, the GARCH model is estimated based on the observations of the past window.

For example, if we choose a forecast window of 48 months, we forecast $\hat{\sigma}_t^2$ based on the GARCH parameters estimated using the returns of months $[t-48, t-1]$. This forecasting is then carried out for the full sample $t = 49, \ldots, T$.

For forecasting with a window size in days, for any month t s.t. there at least W_d days of data before the first day d of month t, we estimate the GRARCH parameters using the returns of days $[d-W_m, d-1]$ to forecast $\hat{\sigma}_d^2$. We then set $\hat{\sigma}_t^2 = \hat{\sigma}_d^2$.

For any month for which there is not enough data available to estimate a full window, we use weights 1/N instead.

For comparison, we also used the full sample to create a forward looking estimate of the GARCH parameters. This is of course not a realistic model as we are interested in forecasts. Still, the clairvoyant parameters are used to calculate the monthly variances to provide a comparison.

After estimating the returns in these ways, we are able to compute comparison measures, which are presented in Section 5.1.

Table 4: GARCH results

	Ann. Return (%)	Sharpe	Sortino	Skewness
$W_m = 240$	**3.65**	**0.92**	**1.41**	0.13
$W_m = 120$	3.09	0.82	1.28	0.21
$W_m = 48$	3.10	0.81	1.24	0.29
$W_d = 5000$	**3.65**	0.89	1.28	0.11
$W_d = 2500$	3.23	0.81	1.19	0.06
$W_d = 1000$	3.20	0.85	1.27	0.13
Full sample	3.45	0.90	1.38	0.47

Interestingly enough, the 240 month window performs much better than the 120 and 48 month windows, based on each performance measure, while the sizes of the day based windows matter less. It even provides better results than the full sample estimation, which uses future returns to estimate today's parameters. This suggests that behavior modeled by GARCH does not stay constant over time.

Furthermore, the 240 month rolling-window forecast is also the only GARCH model that performs better in both annualized returns and Sortino ratio than the simple volatility managed model and the 1/N portfolio we benchmark against. In other words, not every one-step forecasting approach manages to perform better than the benchmark models. This shows that the relatively easy benchmark models may perform surprisingly well.

Based on these results, we suspect that a 240 month window size will deliver the most optimal set of risk metrics for the GARCH model. For this reason, and also to keep our analysis somewhat organized, we will use the $W = 240$ months forecast window to check the performance of the models.

5.2 Model Performance

We include multiple findings in Section 5.2. First, we provide results for the models that were coded ourselves in Matlab. Then, we provide results of models that make use of the model estimators from the Mathworks Econometrics toolbox and the Oxford MFE toolbox. We provide the measures that were also used for the benchmark models to make a broad comparison for the different models.

Table 5: 240 Month Window Models

Model Name	Ann. Return (%)	Sharpe	Sortino	Skewness
Custom Code				
Garch - Gaussian	3.65	**0.92**	1.41	0.13
GAS - Gaussian	3.78	**0.92**	**1.45**	0.39
GAS - Student's t	3.38	0.66	0.91	-0.23
Matworks Econometrics toolbox				
Garch	3.66	0.90	1.40	0.30
Egarch - Gaussian	3.57	0.79	1.13	0.74
Garch - Student's t	3.67	0.90	1.40	0.33
Egarch - Student's t	3.55	0.78	1.13	0.72
Oxford MFE Toolbox				
Garch	3.76	**0.92**	1.37	0.12
EGarch - Gaussian	3.69	0.89	1.42	0.55
Garch - Student's t	3.65	0.91	1.40	0.25
EGarch - Student's t	3.72	0.90	1.40	0.37
Garch - GED	3.70	**0.92**	1.41	0.27
EGarch - GED	3.70	0.90	1.40	0.43
DCC	3.61	**0.99**	**1.48**	**0.09**
Benchmark Models				
1/N	**3.85**	0.86	1.14	-0.29
Simple Vol.	3.25	0.87	1.36	0.29

We first take a look at the uni-variate models. Our analysis shows that the GAS-Normal model has the highest annualized return at 3.78%, followed by the Garch model from Oxford MFE toolbox at 3.76%. Both models have the highes Sharpe ratios (0.92) among the univariate models, respectively, indicating that they generate higher returns whilst maintaining a relatively lower volatility compared to other models.

The Sortino ratio measures the downside risk of a portfolio. The GAS-Normal model, again, provides the best performance with a ratio of 1.45, implying that the model has a more accurate downside risk assessment. Additionally, the EGarch model from Oxford MFE toolbox has a good Sortino ratio, indicating that it can efficiently handle downside risk.

It is promising to observe that the modelling of inter-dependencies and time-dynamics across different factors by means of a multivariate DCC model, yields the best Sharpe and Sortino ratios. The Sharpe ratio of 0.99 and Sortino ratio of 1.48 indicate superior risk-adjusted returns compared to all other implemented models.

Finally, the skewness of a model tells us about the presence and the degree of asymmetry in the returns. We can see that most models have positive skewness values, which implies that they tend to generate positive returns more frequently than negative returns. The DCC model has the smallest absolute value of skewness which means it indicates a more symmetric distribution.

Considering the overall performance of the models, the DCC model seems to be the most promising due to its superior performance for all risk-adjusted portfolio measures.

6 Conclusion

For an investor that wants to minimize their risk exposure for some target expected return, almost all introduced models outperform the 1/N benchmark model. Also, most models were able to deliver better Sharpe and Sortino ratios than the simple volatility managed benchmark models. It is interesting to note that the 1/N portfolio still yields the highest annual returns, which could be of interest to investors with higher risk-tolerances. But, for most investors who care about their exposure to risk, based on the risk-adjusted metrics, we are able to improve portfolio performance by incorporating econometric models that predict volatility of factors over time.

Further research could focus on comparing other volatility forecasting models, beyond the ones implemented in the paper. This could involve models that address different characteristics of financial data, such as for example regime-switching behaviour and seasonality.

Another suggestion is to compare out-of-sample forecasting performance of the implemented models for an even more nuanced comparison, as differing results may arise.

One interesting result from Section 5.1 is that a 20 year rolling window is only optimal if GARCH is estimated based on monthly returns. When using daily returns it barely gives an improvement over shorter window sizes. Future research could explore the reason for this difference and perhaps improve upon GARCH on this dimension.

References

D. Alberg, H. Shalit, and R. Yosef. Estimating stock market volatility using asymmetric garch models. *Applied Financial Economics*, 18(15):1201–1208, 2008.

T. Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3): 307–327, 1986.

T. Bollerslev. Modelling the coherence in short-run nominal exchange rates: A multivariate generalized arch model. *The Review of Economics and Statistics*, 72(3):498–505, 1990.

M. Caporin and M. McAleer. Robust ranking of multivariate garch models by problem dimension. *Computational Statistics & Data Analysis*, 76(1):172–185, 2014.

D. Creal, S. J. Koopman, and A. Lucas. Generalized autoregressive score models with applications. *Journal of Applied Econometrics*, 28(5):777–795, 2013.

V. DeMiguel and L. Garlappi. Optimal versus naive diversification: How inefficient is the 1/n portfolio strategy? *The review of Financial studies*, 22(5):1915–1953, 2009.

V. DeMiguel, Y. Plyakha, R. Uppal, and G. Vilkov. Improving portfolio selection using option-implied volatility and skewness. *Journal of Financial and Quantitative Analysis*, 48(6):1813–1845, 2013.

Z. Ding, C. W. Granger, and R. F. Engle. A long memory property of stock market returns and a new model. *Journal of empirical finance*, 1(1):83–106, 1993.

R. Engle. Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3):339–350, 2002.

R. F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. *Econometrica: Journal of the econometric society*, pages 987–1007, 1982.

E. F. Fama and K. R. French. Multifactor explanations of asset pricing anomalies. *The Journal of Finance*, 51(1):55–84, 1996.

E. F. Fama and K. R. French. A five-factor asset pricing model. *Journal of Financial Economics*, 116 (1):1–22, 2015.

French Data Library. Kenneth r. french - data library, 2023. URL `https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html`. Retrieved on 2023-05-30.

L. R. Glosten, R. Jagannathan, and D. E. Runkle. On the relation between the expected value and the volatility of the nominal excess return on stocks. *The journal of finance*, 48(5):1779–1801, 1993.

C. M. Hafner and O. Reznikova. On the estimation of dynamic conditional correlation models. *Computational Statistics & Data Analysis*, 56(11):3533–3545, 2012.

H. Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.

H. M. Markowitz. Foundations of portfolio theory. *The Journal of Finance*, 46(2):469–477, 1991.

A. Moreira and T. Muir. Volatility-managed portfolios. *The Journal of Finance*, 72(4):1611–1644, 2017.

D. B. Nelson. Conditional heteroskedasticity in asset returns: A new approach. *Econometrica: Journal of the Econometric Society*, pages 347–370, 1991.

D. B. Nugroho, D. Kurniawati, L. P. Panjaitan, Z. Kholil, B. Susanto1, and L. R. Sasongko1. Empirical performance of garch, garch-m, gjr-garch and log-garch models for returns volatility. *Journal of Physics: Conference Series*, 1307(1), 2019.

M. J. Rodríguez and E. Ruiz. Revisiting several popular garch models with leverage effect: Differences and similarities. *Journal of Financial Econometrics*, 10(4):637–668, 2012.